Amirali Imanipanah

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Professional Summary

Electrical Engineering (Electronics) graduate specializing in **STM32-based Embedded Systems**, **Autonomous Robotics**, and **PCB Design**. Experienced in ROS 2, hardware–software integration, and FPGA-based Digital Design.

Education

University of Isfahan Isfahan, Iran

Bachelor of Science in Electrical Engineering (Electronics) Sep 2022 – Sep 2026

- **Overall GPA:** 16.80 / 20.00
- **Key Courses:** Microcontrollers, Digital Systems Design, Computer Architecture, Signals & Systems.

Research Interests: Autonomous Robotics, Embedded Control, Edge AI, FPGA-based Digital Systems.

Technical Skills

Embedded & Firmware: STM32 (F1/F4/H7), ESP32, RP2040, AVR/STM8, C/C++, FreeRTOS, DMA, TouchGFX, UART, SPI, I2C, ADC, PWM.

Robotics & Linux: ROS 2, Raspberry Pi 5, Embedded Linux, Python.

Hardware & PCB: Schematic/PCB Design, Altium Designer, KiCad, PSpice, Proteus, SMD Assembly.

Digital Design & Tools: FPGA/Verilog, MATLAB, Git, Linux CLI, SolidWorks.

Certifications & Training

IEEE Student Branch: Professional PCB Design (Advanced, 2024); Introductory PCB Design (2023).
Faradars: Four-Layer PCB Design (Altium Designer); Practical Four-Layer FPGA Board Design.

Teaching Experience

Teaching Assistant (TA) *University of Isfahan*

- **Spring 2026:** Microcontrollers (AVR/STM32, 80+ students); Electronics I (transistor circuits).
- **Fall 2025:** Electric Circuits I (circuit analysis); Fundamentals of Programming (C++ and debugging).

Selected Projects

ROS 2-Based Autonomous Mobile Robot *B.Sc. Thesis*

- Architected distributed robotics software across **Raspberry Pi 5** (ROS 2), **RP2040** (closed-loop motor control/safety), and a remote vision-processing node.
- Implemented CRC-protected Pico–Pi communication, wheel odometry, LiDAR aggregation, and embedded **Neuro-PID** inference with offline training scripts.
- Verified host-side logic with **18/18 behavior** and **10/10 fail-safe** checks; benchmarked application-layer offloading through software emulation.
- Designed **five KiCad PCB modules** for power distribution, motor power, Pico interfacing, Pi protection, and thermal management.

STM32-Based HMI & Embedded Instrument Platform *Independent Project*

- **V1 – STM32F407:** Developed a graphical TFT/touch HMI with real-time graphing and matrix-processing functionality.
- **V2 – STM32H723:** Designing a four-layer instrument with a shared FMC bus for SDRAM and TFT, capacitive touch, ESP32-C6, and expansion I/O.
- Analyzed FMC bus timing, signal allocation, and PCB placement constraints for the upgraded architecture; hardware validation pending.

Zero-DSP FPGA Multi-Sensor Accelerator *Digital Systems Project*

- Designed multiplierless Verilog processing for LiDAR/IMU: CORDIC coordinate transforms, adaptive shift-based Kalman filtering, and 4-tap distributed-arithmetic FIR.
- Compared **MATLAB HDL Coder** variants with a custom **13-stage CORDIC** pipeline; tool-reported **115.399 MHz Fmax** on Spartan-6 with zero DSP slices.

Languages

English: Professional working proficiency. **Persian:** Native.